

Kuan-Chen Chen

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EDUCATION

University of Florida

Master of Science in Applied Data Science

Gainesville, FL

8/2024 - 5/2026

National Sun Yat-Sen University

Bachelor of Science in Applied Mathematics

Kaohsiung, Taiwan

9/2019 - 6/2023

Honors & Awards: The Valedictorian of Department of Applied Mathematics, Distinguished Service Award

PROJECT EXPERIENCE

cAIuldron (Recipe Generation LLM) [Ckcinnabar/cAIuldron](https://github.com/Ckcinnabar/cAIuldron)

Gainesville, FL

- Built v1 end-to-end ML pipeline integrating CLIP (ViT-Base) and DETR (ResNet-50) for multi-ingredient detection (88% F1, 525+ classes); fine-tuned Llama 3.2 1B via QLoRA, training only 0.23% of parameters (2.8M) to achieve 92/100 recipe quality on 7,913 curated RecipeNLG recipes.
- Rebuilt as cloud-native v2 using LangGraph StateGraph orchestration; replaced local GPU models with Groq Llama 4 Scout 17B (vision, ~1s) and Llama 3.3 70B, eliminating 4–6 GB VRAM dependency and upgrading from sequential Python calls to parallel agent execution.
- Integrated ChromaDB RAG + Tavily web search via LangGraph fork/join; delivered 6-tab Gradio UI with 5 concurrent FLUX.1-schnell async image calls and LangSmith observability, cutting pipeline latency from 40–80s (v1) to 10–20s (v2).

Climate Change Innovation Competition, The Ministry of Education (Taiwan)

Taipei, Taiwan

- Built DARTS campus mobility prediction system integrating multi-source APIs (TDX/PTX), enabling automated 24/7 data pipeline and real-time forecasting of pedestrian, bicycle, and bus demand
- Developed hybrid time-series model (ARIMA-LSTM) with data preprocessing and feature engineering, achieving 86.1% accuracy; delivered actionable insights adopted by university, optimizing bus schedules and reaching national finals (Top 121 teams)

WORK EXPERIENCE

Florida Gators Women's Basketball

Gainesville, FL

Data Scientist Intern

8/2025 - Present

- Built an opponent-strength-adjusted player evaluation system (OA-Rating) using XGBoost and SHAP, analyzing 120K+ game-level observations across 5,500+ NCAA players and 350+ teams
- Implemented an Elo-based team rating system ($\rho = 0.83$ vs. NCAA NET) to track real-time team strength and support game preparation
- Developed a production R Shiny dashboard for player comparison and opponent scouting, delivering insights that contributed to team improvement from 16–17 (NET 60) to 18–15 (NET 48)

Bioinformatics Lab, University of Florida

Gainesville, FL

Student Research Assistant

12/2024 - Present

- Designed stacking ensemble algorithm combining XGBoost, LightGBM, and Balanced Random Forest with logistic regression meta-learner to classify 10,486 T-cell receptor CDR3 sequence, improving robustness on highly imbalanced data (1:23) and achieving ROC-AUC 0.876
- Developed biologically-informed data augmentation algorithm using BLOSUM62 substitution matrix, generating conservative sequence variants to expand minority class 10x (432 → 4,622) and mitigate severe class imbalance (1:23 → 1:1.5)
- Engineered 42 physicochemical features from protein sequences and optimized decision thresholds under class imbalance, improving precision to 0.716 for high-confidence, interpretable predictions.

Optimized AI Conference

Atlanta, GA

Volunteer Staff

4/2025, 3/2026

Glory Integrated Marketing Co. Ltd

Taipei, Taiwan

Data Analyst Intern

8/2023 - 12/2023

SKILLS

- Languages: Python, SQL, R | Databases: MySQL, MongoDB, Oracle
- ML/DL: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace Transformers, SHAP
- Web & Viz: Flask, React.js, Gradio, Dash, Shiny, Power BI, Tableau